THE LOCAL COMPLETE INTERSECTION PROPERTY FOR A RING MAP IS LOCAL ON THE TARGET IN THE ZARISKI TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Local complete intersection for a ring map, expressed through the Koszul regularity of the ideal presenting a localisation. used throughout the proof, source lines 8354-8383. Supplies the hypothesis at each localisation and the goal at the original ring.
- Independence of the choice of generators for a Koszul-regular ideal. the statement counted by the sibling card of this lane for Lemma lemma-independence-of-generators at 7711-7755, cited at proof lines 8375-8378. Turns the generating Koszul-regular sequence into the specific sequence obtained from the conormal basis.
- Truncation of a Koszul-regular sequence into a quotient. the statement counted by the sibling card S1-algebra-truncate-koszul-regular at 7679-7709, cited at proof lines 8378-8381. Descends Koszul regularity from the extended presentation to the original one.

Source reference. Original lemma 07D1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9396–9396. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 08JZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10867–10867. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 066A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7929–7929. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 068M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8011–8011. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $R \to S$ be a ring map. Let $g_1, \ldots, g_m \in S$ generate the unit ideal. If each $R \to S_{g_j}$ is a local complete intersection so is $R \to S$.*

Proof. Let $S = R[x_1, \ldots, x_n]/I$ be a presentation. Pick $h_j \in R[x_1, \ldots, x_n]$ mapping to g_j in S. Then $R[x_1, \ldots, x_n, x_{n+1}]/(I, x_{n+1}h_j - 1)$ is a presentation of S_{g_j}. Hence $I_j = (I, x_{n+1}h_j - 1)$ is a Koszul-regular ideal in $R[x_1, \ldots, x_n, x_{n+1}]$. Pick a prime $I \subset \mathfrak{q} \subset R[x_1, \ldots, x_n]$. Then $h_j \notin \mathfrak{q}$ for some j and $\mathfrak{q}_j = (\mathfrak{q}, x_{n+1}h_j - 1)$ is a prime ideal of $V(I_j)$ lying over $\mathfrak{q}$. Pick $f_1, \ldots, f_r \in I$ which map to a basis of $I/I^2 \otimes \kappa(\mathfrak{q})$. Then $x_{n+1}h_j - 1, f_1, \ldots, f_r$ is a sequence of elements of I_j which map to a basis of $I_j \otimes \kappa(\mathfrak{q}_j)$, see Algebra, Lemma 08JZ. By Nakayama's lemma there exists an $h \in R[x_1, \ldots, x_n, x_{n+1}]$ such that $(I_j)_h$ is generated by $x_{n+1}h_j - 1, f_1, \ldots, f_r$. We may also assume that $(I_j)_h$ is generated by a Koszul regular sequence of some length e. Looking at the dimension of $I_j \otimes \kappa(\mathfrak{q}_j)$ we see that $e = r + 1$. Hence by Lemma 066A we see that $x_{n+1}h_j - 1, f_1, \ldots, f_r$ is a Koszul-regular sequence generating $(I_j)_h$ for some $h \in R[x_1, \ldots, x_n, x_{n+1}]$, $h \notin \mathfrak{q}_j$. By Lemma 068M we see that $I_{h'}$ is generated by a Koszul-regular sequence for some $h' \in R[x_1, \ldots, x_n]$, $h' \notin \mathfrak{q}$ as desired. □